Yash Chauhan

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Fourth Year Int. MSc. Physics Major

Experience

RPTU Kaiserslautern - Theoretical Quantum Optics Group, *Summer Intern* (Prof. Michael Fleischhauer) | Kaiserslautern, Germany — *Jul 2025 - Present*

- Working on deriving Effective Hamiltonains for Soliton solutions in a 2D Topological Insulator with interactions.

National Institute for Materials Science - Surface Quantum Phase Materials Group, *Summer Intern* (Prof. Takashi Uchihashi) | Tsukuba, Japan — *May 2025 - June 2025*

- Grew **atomic-layer** Indium **superconductors** on Si substrates via sputtering; studied **2D superconductivity**.
- Performed transport measurements using a **UHV STM + Transport** setup (0-8T, 300mK).
- Investigated **Josephson vortex** formation and dynamics under perpendicular and parallel magnetic fields.

NISER - Laboratory of Novel Magnetism and Spintronics, *Research Volunteer* (Prof. Ajaya Kumar Nayak) | Bhubaneswar, India — *Aug 2024 - Present*

- Currently conducting a literature review and exploring potential Master's thesis projects in the direction of **quantum skyrmions**, **skyrmion-based qubits**, and **topological QC**.
- Initially involved in synthesizing potential **altermagnetic** rare earth compounds via arc-melting and self-flux methods (now paused); currently learning thin film deposition techniques like PLD and sputtering, and exploring theoretical tools such as Quantum ESPRESSO, **OOMMF**, and **ubermag**.

Raman Research Group, Tata Institute of Fundamental Research, *Summer Intern* (Prof. Karthik V. Raman) | Hyderabad, India — *June 2024 - Aug 2024*

- Worked in Fabricating Thin Epitaxial Films of $(\mathbf{Bi_xSb_{1-x}})_\mathbf{2}\mathbf{Te_3}$ topological insulator.
- Acquired hands-on experience in **MBE, XRD, AFM, ICE** cryostat, transport measurements.
- Also explored theoretical condensed matter and many body physics - Topological matter, **Neutral Atom Quantum Simulation** using Rydberg Arrays and Altermagnetism.
- Went back during the winter break (*Dec, 2024*) to acquire hands-on experience in operating **STM**. Studied the surface of KV_3Sb_5, a kagome superconductor.

Quantum Denoising for Brain-Computer Interfaces/Quantum SVM, *Intern* (Dr. Grégoire Cattan) | Remote — *Feb 2024 - Feb 2025*

- Investigated self-supervised learning for **EEG**-based passive Brain-Computer Interfaces (pBCIs)
- Explored **Quantum Autoencoders** for signal denoising.
- Abstract accepted at the **5th International Neuroergonomics Conference**
- Compared Quantum and RBF kernels for the Diagnosis of **Parkinson** using **QSVM**

NISER - Color Spanning Sets, *Research Volunteer* (Dr. Abhishek Sahoo) | Bhubaneshwar, India — *Dec 2023 - May 2024*

- Studied the fixed parameter tractability of some of the **NP-hard** Color Spanning Set problems.

AICH Lab - MDNA, *Research Volunteer* (Prof. Palok Aich) | Bhubaneshwar, India — *June 2023 - Oct 2023*

- Worked on understanding the **HOMO-LUMO** band gap of **metal-doped DNA** structures.
- Performed ab-initio **DFT** calculations using ORCA and modelled the structures on Avogadro.

Indian Statistical Institute, *Student Intern* (Dr. Ramij Rahaman) | Kolkata, India — *Mar 2023*

- Learnt about several **QKD** protocols - implementations and attacks.
- Worked on understanding the security proof of the **BB84** protocol.

National Physical Laboratory, *Winter Intern* (Dr. H.K. Singh) | New Delhi, India — *Dec 2022*

- Learnt about **Thin Film** depositions using the **RF Magnetron Sputtering** System.
- Participated in the fabrication of **PCMO** on substrates of Lanthanum Aluminate and Strontium Titanate

Education

8.45/10 **Integrated MSc. in Physics**, *National Institute of Science Education and Research* | Bhubaneshwar, India — *2022-27*

92.30 % **Std. XII, CBSE**, *Career Point Gurukul* | Kota, India — *2022*

94.50 % **Std. X, ICSE**, *Sharda Mandir School* | Goa, India — *2020*

Achievements: DAE DISHA Scholarship | WOMANIUM Quantum Scholarship | Tensorflow Education Stipend | Travel Award - 5th International Neuroergonomics Conference

Conference: National Quantum Science and Technology Symposium (QETCI)

Research Interest: Topological Quantum Matter, Spintronics, Altermagnetism, Many-Body Physics, Quantum Algorithms, Quantum Information, Systems Biology, Mathematical Modelling

Skills

Experimental	MBE, STM, AFM, XRD, ICE Cryostat, Hall Bar Measurements, Transport Measurements, Wire Bonding/Scratching, Arc-Melting, Self-Flux Method, Sputtering
Programming	Python, C/C++, C#, Javascript, Git, Bash, LaTeX, HTML, Julia, CSS, Java
Software	Linux, Github, ORCA, Avogadro, LabVIEW, Gwyddion, ORIGIN, OOMMF | qiskit, cirq, pennylane, tensorFlow, bloqade.jl, yao.jl, Data Analysis
Certifications	• IBM Certified Associate Developer - Quantum Computation using Qiskit v0.2X • DeepLearning.AI Tensorflow Developer Professional Certificate • Qiskit Global Summer School 2023 | Qbronze, QNickel, QSilver, QMercurcy - QWorld • TCS Quantum Challenge 2023

Projects

Understanding Algorithmic Learning Through Transformer Shortcuts *April 2025*

NISER, CS461 Advanced Machine Learning: Prof. Anup Kumar Bhattacharya

- Based on the paper *"Transformers Learn Shortcuts to Automata"* by Bingbin Liu et al., conducted a detailed study of **how shallow Transformers simulate automata-like logic**.
- Explored shortcut solutions that enable Transformers to bypass sequential processing using parallel architectures.
- Investigated theoretical tools like Krohn–Rhodes decomposition and circuit complexity, and evaluated empirical robustness under varying training conditions.

A Quantum Approach to Option Pricing *Nov 2024*

NISER, P346 Computational Physics: Prof. Colin Benjamin

- Replicated key results from the seminal paper "Option Pricing Using Quantum Computers" by Stamatopoulos et al., showcasing the application of **Quantum Amplitude Estimation** for option pricing.
- Designed and implemented quantum circuits to price vanilla, basket, and path-dependent options using Qiskit, analyzing potential quadratic speed-up over classical Monte Carlo methods.
- Investigated challenges in applying quantum algorithms to real-world financial problems, emphasizing hardware requirements and algorithm scalability for quantum finance applications.

Optimising Fleet Allocation *Nov 2023 - Mar 2024*

TCS Quantum Challenge 2023, Remote

- Conducted background research on the **Airline Scheduling** problem and performed preliminary experimental validation as part of Phase I.
- Worked with a team of four members accompanied by technical and academic advisors on **Quantum Optimization** using **Amazon Braket**.
- Reached the Final stage and delivered a presentation to a panel consisting of eminent academicians from Oxford and CEOs of several airline companies.

Publications

Preprints

- **Chauhan, Y.** (2024, March). Bhubaneswar through the lens of Invisible Cities: Exploring geometrical narratives. `https://doi.org/10.31235/osf.io/fqz9g`

Published

- Cattan, G., Olmos, J., **Chauhan, Y.** et al. Parkinson disease classification: a comparison of quantum and RBF kernels using support vector machine. Quantum Mach. Intell. 7, 64 (2025). https://doi.org/10.1007/s42484-025-00286-0

- Ahmed Mostafa, **Yash Chauhan**, Wasif Ahmed, Grégoire Cattan, Marius Klug. Quantum Denoising for BCI: A Study. 5th International Neuroergonomics Conference, University of Bordeaux, Jul 2024, Bordeaux, France. pp.494-499. (hal-05020004)

WORKSHOPS/EVENTS

13th Annual International Hackathon for Social Good *Apr 2025*

New York University, Abu Dhabi (NYAUD)

- Developed a proof-of-concept pipeline for cancer risk minimization using radiotherapy, integrating beam angle optimization in hardware setups.
- Designed a frontend system for cancer detection from radiological data as part of Team QurAI.

QUIKSTART x QML *Feb 2025*

RPTU Kaiserslautern, Germany

- Selected to attend a winter school on **Quantum Machine Learning** and received a grant of **€1590** for participation.
- Presented a poster titled *Quantum Denoising for BCI* during the school.

Extra-curricular

Chess Club	Organised intra-college events, friendly matches and promoting chess
Tennis Club	Represented the college in tournaments
Physics Club	Started a subgroup focused on mathematical physics

Additional Coursework

2023 **Quantum Algorithms and Cryptography**, IIT Madras, NPTEL
2023 **Graph Theory**, IISER Pune, NPTEL
2023 **Cryptography I**, Stanford Online, Coursera
2024 **Metric Spaces**, NISER, In Person
2025 **Introduction to Manifolds**, NISER, In Person

Languages

English Professional proficiency
Hindi Native proficiency
German Currently Learning (Beginner)